Md. Farhan Mahtab

Dhaka, Bangladesh
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Education

Bangladesh University of Engineering and Technology April, 2019 – June, 2024
B.Sc in Computer Science and Engineering | CGPA: 3.82/4.00 *Dhaka, Bangladesh*

Research Experience

Systematic Vulnerability Analysis of Matter SDK August, 2026 – Present
University of Texas Dallas *Dallas, Texas, USA*

- Investigating security vulnerabilities in the Matter SDK through formal models generated from system specifications using large language models (LLMs).
- Exploring an LLM-assisted formal modeling approach for systematically analyzing the security properties of software specifications.
- Working at the intersection of LLMs, formal methods, IoT security, and automated vulnerability analysis

FedCrime: A Federated Learning Based Crime Prediction Model July, 2023 – May, 2024
Bangladesh University of Engineering and Technology *Dhaka, Banagladesh*

- Undergraduate Thesis Supervised by Dr. Tanzima Hashem
- Developed a privacy-preserving crime prediction approach using federated learning over private crime data
- Investigated how distributed learning can improve predictive modeling while avoiding direct sharing of sensitive data
- Explored the application of federated learning to privacy-sensitive predictive analytics

Myers-Briggs Personality Type Identification Using LLM October,2023 – February, 2024
Bangladesh University of Engineering and Technology *Dhaka, Banagladesh*

- Fine-tuned a large language model to generate embeddings from social media text
- Developed a classification model using LLM-derived text embeddings to identify the INTJ personality type
- Investigated the use of LLM representations for downstream classification of personality-related textual information

Job Experience

Software Engineer June, 2024 – Present
Samsung R&D Institute Bangladesh *Dhaka, Bangladesh*

- Developed and maintained software for the Tizen Developer Interface and Tizen developer ecosystem
- Optimized the Tizen emulator based on the open-source QEMU project, focusing on emulator performance
- Developed Rust-based GUI tools for the Tizen SDK
- Developed an AI agent for the Tizen SDK, combining software development tooling with AI-based capabilities
- Gained hands-on experience in systems software, emulation, developer tooling, operating-system ecosystems, and AI-assisted software development

Intern May,2023 - June, 2023
ERA-InfoTech Limited *Dhaka, Bangladesh*

- Developed an unsupervised machine learning model for classification of bank transactions
- Applied machine learning techniques to financial transaction data for automated categorization

Research & Technical Areas

Trustworthy AI

- LLM-assisted Security Analysis · LLMs · Formal Modeling · Vulnerability Analysis

Machine Learning

- Federated Learning · Privacy-Preserving ML · Representation Learning · Text Classification · Unsupervised Learning

Systems & Security

- IoT Security · Matter SDK · Tizen · QEMU · Emulator Optimization · Systems Software

Development

- Rust · Software Engineering · SDK & Developer Tooling · AI Agents

Awards & Honors

Icon Engineer
Samsung R&D Institute Bangladesh *2026*

RISE Student Research Grant
Bangladesh University of Engineering and Technology *2023*

Dean's Award
Bangladesh University of Engineering and Technology *2021*